

DES exposure and the aging woman: mothers and daughters.

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Diethylstilbestrol (DES), the first orally active artificial estrogen ever developed, was prescribed to several million pregnant women during the 1940s through the 1960s in the mistaken belief that it reduced the risk of miscarriage. In 1971, the US Food and Drug Administration contraindicated its use in pregnancy when DES was associated with the development of vaginal clear cell adenocarcinoma (CCA) in daughters exposed in utero. In daughters whose mothers took DES during pregnancy, the drug has been associated with congenital malformations of the reproductive tract, fertility problems, a possible increased risk of cervical carcinoma in situ, and a presumed lifetime risk of vaginal and cervical CCA. DES mothers have an increased risk of breast cancer (RR = 1.3). DES sons have an increased prevalence of urogenital anomalies, and a possible increased risk of testicular cancer.

Management of pelvic floor disorder in a diethylstilbestrol daughter.

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Between 1947 and 1971, the synthetic estrogen diethylstilbestrol (DES) was prescribed to pregnant women to prevent spontaneous abortions. Many studies have demonstrated adverse consequences of prenatal DES exposure. This report describes a DES daughter treated for uterine cancer who presented with a foreshortened vagina of 4 cm and stage III pelvic organ prolapse. She underwent successful vaginal native tissue reconstructive surgery with uterosacral colpopexy and anterior and posterior repair to lengthen her vagina and treat her prolapse. She then developed stress incontinence and underwent periurethral bulking 5 months later. She has been followed for 2 years with no recurrence of prolapse or incontinence.

The history of DES, lessons to be learned.

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Since the 1940s, diethylstilbestrol (DES) has been used by millions of pregnant women to prevent miscarriages and many other disorders in pregnancy. In 1971, it became clear that this apparently innocent treatment proved to be a time bomb for the infants exposed to DES during the first trimester of pregnancy. DES is now associated with an increased risk of breast cancer, clear cell adenocarcinoma (CCAC) of the vagina and cervix, and reproductive anomalies. This article summarises the potential long-term health implications of DES on the mother, DES daughters and DES sons, and the possible side effects on the third generation. Health care professionals have to know the history of DES to prevent future disasters with drugs prescribed.

[Administration of diethylstilbestrol during pregnancy, a public health problem].

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The analysis of the health risks associated with the administration of diethylstilbestrol (DES) during

pregnancy (for the prevention or treatment of threatened abortions) has been largely published. Concerning mothers, a relationship between DES exposure during pregnancy and risk of cancer is unproved. However, existing studies are sufficient cause for serious concern over drug's carcinogenic potential, and further follow-up studies are required. Concerning daughters, a clear association between in utero exposure to DES and clear cell adenocarcinoma of the vagina or cervix is established (incidence between 0.14 and 1.4 per 1000 through age 24). The risk for squamous cell cancer of the vagina and cervix does not seem to be increased. Cervico-vaginal adenosis is frequent (20% to 60% of exposed subjects). This is not a pre-cancerous lesion, its spontaneous evolution is towards regression. No treatment is prescribed. Morphological changes of the genital tract have been described, their consequences on fertility and pregnancy are not clear. Concerning sons, an excess of genital abnormalities (especially of the epididymis and undescended testis) has been reported, but information on the fertility implications of these findings is not available. There is no evidence of an increased risk of testicular cancer. The analysis of all these informations should allow to bring up a policy to take into account these risks in the population.

Diethylstilbestrol induces arrest of rat vascular smooth muscle cell cycle progression through downregulation of cyclin D1 and cyclin E.

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Cardiovascular disease is associated with a multitude of pathophysiologic conditions, including vascular smooth muscle cell (VSMC) proliferation in response to vessel injury. Diethylstilbestrol (DES) was previously prescribed for at-risk pregnancies to prevent abortion, miscarriage, and premature labor. Our aim in this study was to elucidate the effects and molecular mechanism of DES on proliferation and cell cycle progression of platelet-derived growth factor (PDGF)-BB-stimulated rat aortic VSMCs. Treating the cells with DES (1-7 μ M) dramatically inhibited cell proliferation in a dose-dependent manner without any cytotoxic effects. In addition, DES blocked cell cycle progression from PDGF-BB-stimulated cells, which we found was related to down-regulation of the cell cycle regulatory factors, cyclin D1, and cyclin E. Our data demonstrate that DES inhibits rat aortic VSMC proliferation and cell cycle progression through regulation of cell cycle-related proteins. Therefore, our observations may explain, in part, the mechanistic basis underlying the therapeutic effects of DES in cardiovascular disease.

[Prenatal diethylstilbestrol exposure and breast cancer].

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Diethylstilbestrol (DES) is a synthetic estrogen prescribed to prevent miscarriages until 1977. Its role in the development of vaginal adenocarcinoma and cervical dysplasia is known and screening codified. Recent cohort studies show that exposure to DES in utero increases breast cancer risk especially after 40 years. This article reports the observation of a breast cancer of exceptional gravity in a patient exposed to DES in utero. It details the risk of breast cancer for "DES daughters" and support possible screening modalities.

Effects of in utero exposure to DES on male progeny.

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Diethylstilbestrol (DES) was prescribed for almost 30 years by physicians who believed the nonsteroidal synthetic estrogen was beneficial in both the treatment of threatened abortions and the prevention of spontaneous abortions. Toxemia, premature delivery, postmaturity, and stillbirth were said to be significantly reduced in patients who were treated with DES. The consequences of in utero exposure to DES on female progeny have been widely publicized. However, male progeny also suffer the effects of in utero exposure to DES. These effects include epididymal cysts, small penile size, and cryptorchidism. Altered reproductive capacity is suggested by diminished sperm counts and sperm penetration assays.

Reversible inhibition of four important human liver cytochrome P450 enzymes by diethylstilbestrol.

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Diethylstilbestrol (DES), a synthetic estrogen clinically used to treat threatened abortion between the 1940s and the 1970s, has been restricted to treat certain cases of prostatic and breast cancer due to its adverse drug responses such as teratogenicity and carcinogenicity. Some reports have demonstrated that the addition of DES to docetaxel could modify tubulin composition and improve response of prostate cancer to chemotherapy. Given that DES might be co-administered with other drugs such as docetaxel, the present study focused on CYP-based drug-drug interaction (DDI). In vitro inhibitory effects of DES on CYP isoforms were investigated, and the results showed that DES could competitively inhibit CYP3A4, CYP2C8, CYP2C9 and CYP2E1. The inhibition constants (K_i) were calculated to be 4.4 μM , 3.0 μM , 5.0 μM and 8.0 μM for CYP3A4, CYP2C9, CYP2E1 and CYP2C8, respectively. Based on peak serum DES level after drip infusion of 500 mg of fosfestrol (DES diphosphate) in patients, $[I]/K_i$ was calculated to be 4.3, 6.2, 3.7 and 2.3 for CYP3A4, CYP2C9, CYP2E1 and CYP2C8, which suggested that DES was likely to induce in vivo DDI through inhibition of these four major CYP isoforms. These results collectively demonstrate that adverse drug responses might exist when DES is co-administered with other drugs.

Factor XI inhibitors - Rising stars in anti-thrombotic therapy?

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The "holy grail" of preventing and treating thrombosis and thromboembolism would be a drug that was highly effective (preventing clots) and at the same time had a low risk of bleeding. From a hemostasiological perspective, the inhibition of factor XI represents a promising target because a reduced level of factor XI protects against thrombosis without significantly increasing the risk of spontaneous bleeding. Currently, three different classes of drugs of factor XI-inhibition are tested. These are (1) monoclonal antibodies (mAbs), (2) so-called synthetic, small molecules and (3) antisense oligonucleotides (ASOs). This article provides a narrative overview of the current status of studies on all three classes of drugs. Tests with mAbs have been conducted primarily in DVT prevention after knee replacement surgery. One large phase 3 study is testing the mAbs Abelacimab in patients with atrial fibrillation. The synthetic, small molecules Asundexian and Milvexian are tested in several phase 3 trials, mainly in patients with non-cardioembolic ischemic stroke. Results can be expected in the coming years.

Clinical testing of ASOs to inhibit factor XI are still in their infancies.

Therapeutic Potential of FXI Inhibitors: Hype or Hope?

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Significant advancements have shaped the landscape of anticoagulant therapy in the past two decades, including the introduction of direct oral anticoagulants (DOACs), characterized by favorable safety and efficacy profiles and reduced drug-to-drug or food interaction resulting in excellent patient compliance. However, residual concerns still exist with standard-of-care anticoagulant therapy, including the inability to use DOACs in several clinical settings and the need to further reduce the risk of bleeding. Recent improvements in the understanding of the mechanisms behind thrombus formation have led to the awareness that the intrinsic pathway of the coagulation cascade may play an important role in pathological thrombosis, but not in hemostasis. This has represented the rationale for targeting this pathway with factor XI (FXI) inhibitors, with the aim of uncoupling hemostasis and thrombosis. Clinical evidence from patients with FXI deficiency further supports this concept. A number of compounds with different mechanisms of action have been developed to target FXI (i.e., asundexian, abelacimab, lonis-FXIRx, milvexian, osocimab, and Xisomab 3G). To date, the majority of available trials have not gone beyond completion of phase 2 and results are conflictive making it difficult to appraise the clinical benefit of these compounds in the different clinical settings where they have been tested (i.e., atrial fibrillation, acute ischemic stroke, acute myocardial infarction, end-stage renal disease, total knee arthroplasty). Moreover, the largest phase 3 randomized trial designed to test the efficacy of asundexian over apixaban in patients with atrial fibrillation, the OCEANIC-AF, has been prematurely stopped as a result of the inferior efficacy of asundexian. In this review we discuss the pharmacological properties and available evidence generated thus far for factor XI inhibitors, providing a perspective on the current state of these drugs.
